
Customer Segmentation with K-Means

Segmenting 3,900 retail customers, and testing whether the segments are real

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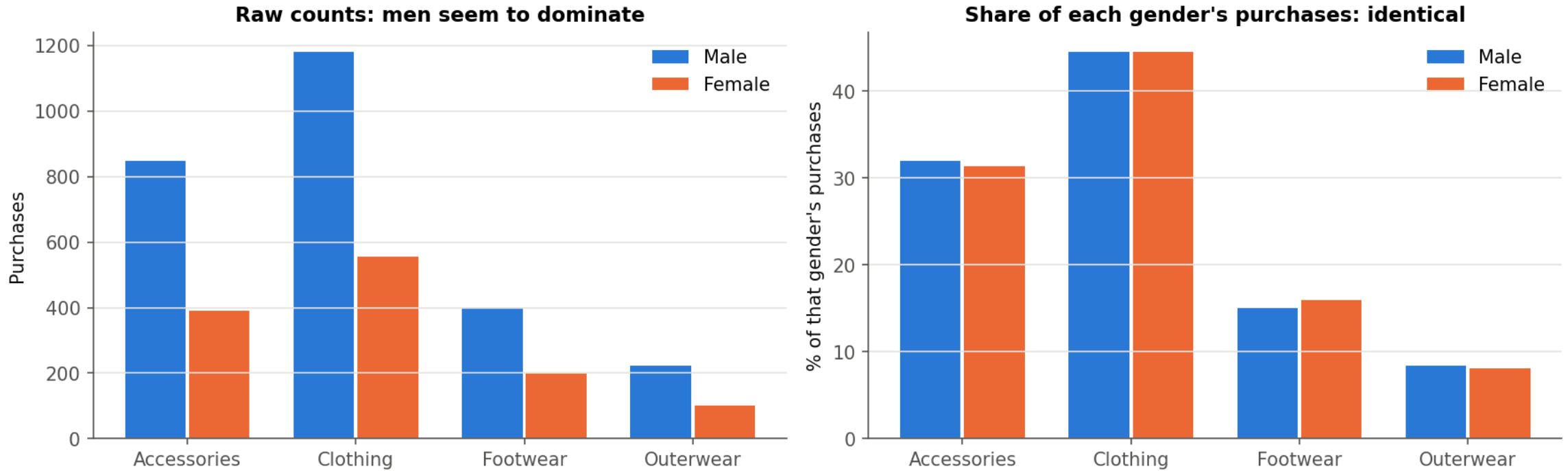
September 2026 | github.com/RatanaSovann/customer-segmentation

Who are our customers, and how should we target them?

Goal: group customers into segments a marketing team can act on.

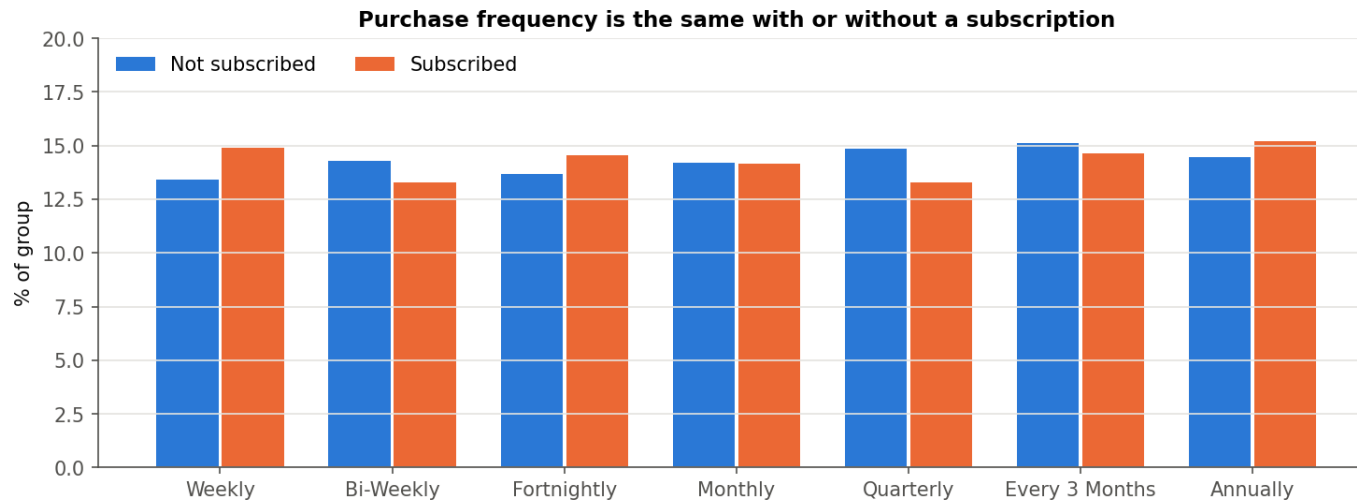
- **Data:** Customer Shopping Trends (Kaggle, synthetic), 3,900 customers, one purchase each, 18 columns
- **Demographics:** age 18-70, gender (68% men), US state
- **Purchase:** 25 products in 4 categories, USD 20-100, season, size, colour
- **Behaviour:** review rating, previous purchases, frequency, subscription (27%), discounts
- **Clean:** no missing values or duplicates

Raw counts mislead: men 'dominate' only because 68% are men

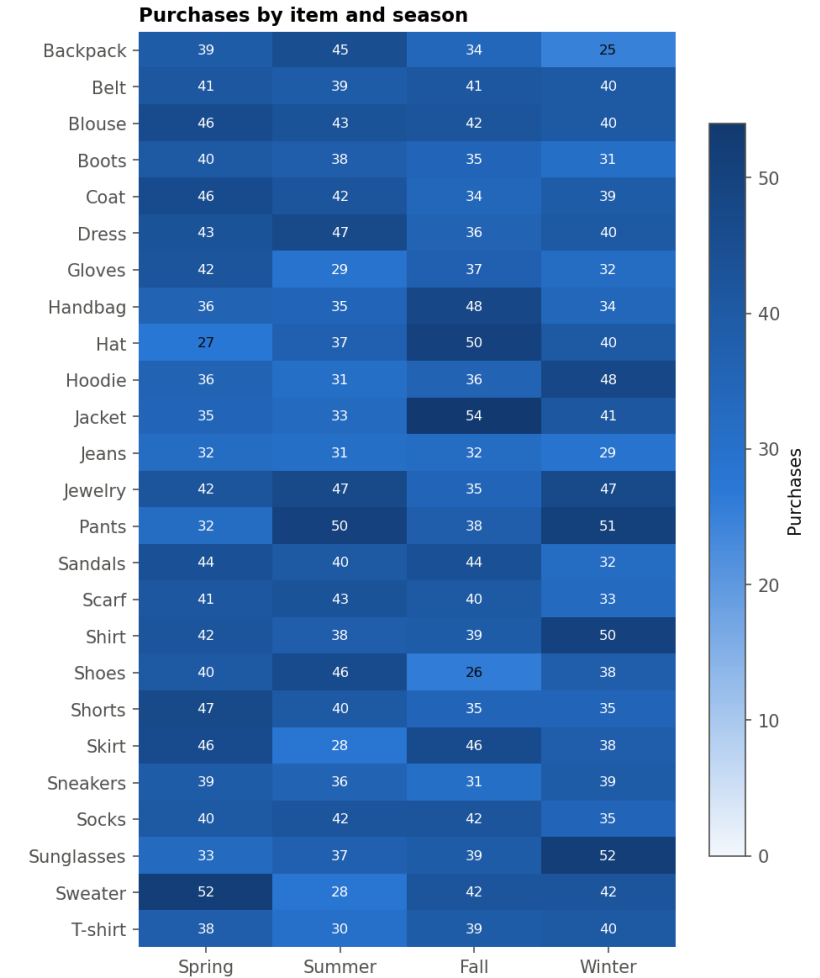


Within each gender, the category mix is identical (chi-square $p = 0.90$).

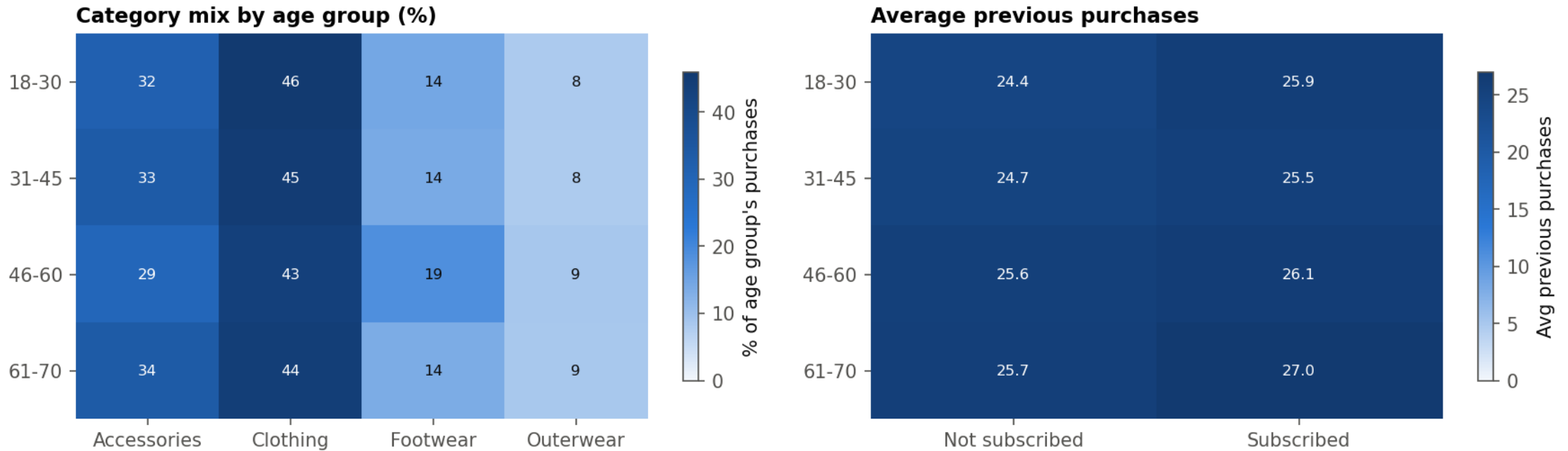
Subscriptions and seasons don't change what people buy



- Same purchase frequency with or without a subscription ($p = 0.69$)
- Item x season counts of 25-54 per cell are consistent with chance ($p = 0.29$)
- All 25 products sell 124-171 units



Age and spend level barely move the needle



- Only borderline difference: 46-60 year-olds buy more footwear (19% vs 14%, $p = 0.053$)
- Subscribers have slightly more previous purchases in every age group (+1 to +1.5)
- Purchase frequency is identical across spend levels ($p = 0.96$)

No test is significant at 5%. Demographics don't predict behaviour in this data.

Approach: K-Means, plus checks that it found something real

1

Features

Age, purchase amount, previous purchases, review rating, standardised. Dollars and counts kept separate.

2

Choose k

Elbow + silhouette for $k = 2-8$, compared with a uniform-noise baseline

3

Profile

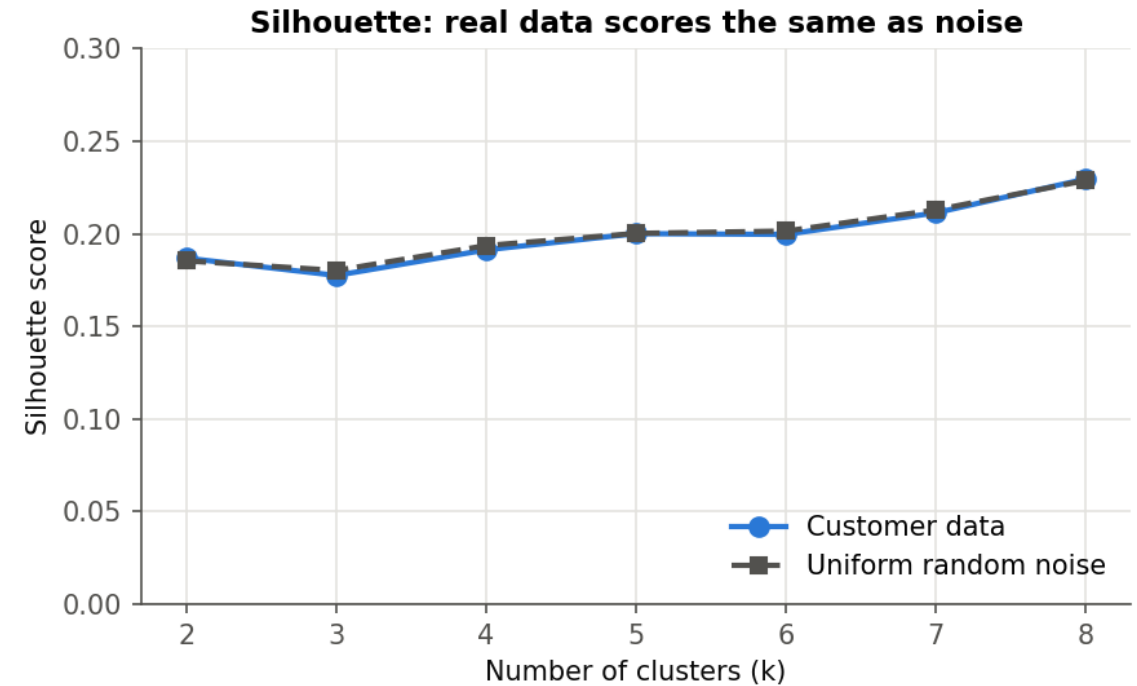
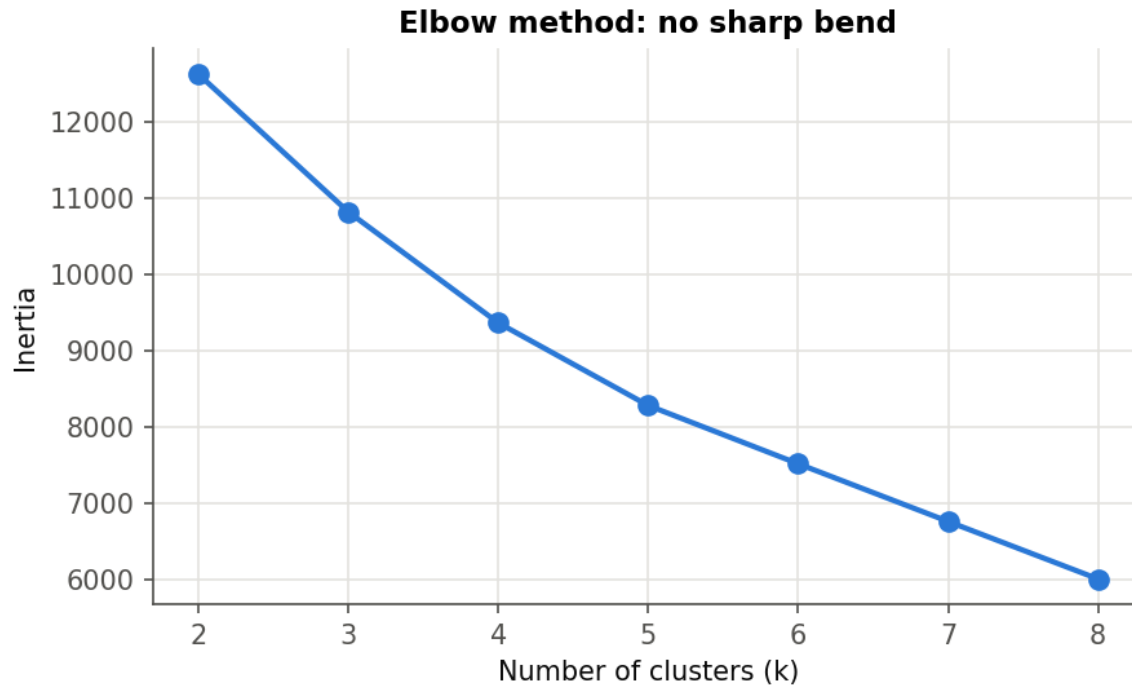
Centroid heatmap, PCA view, named segments, recommended actions

4

Validate

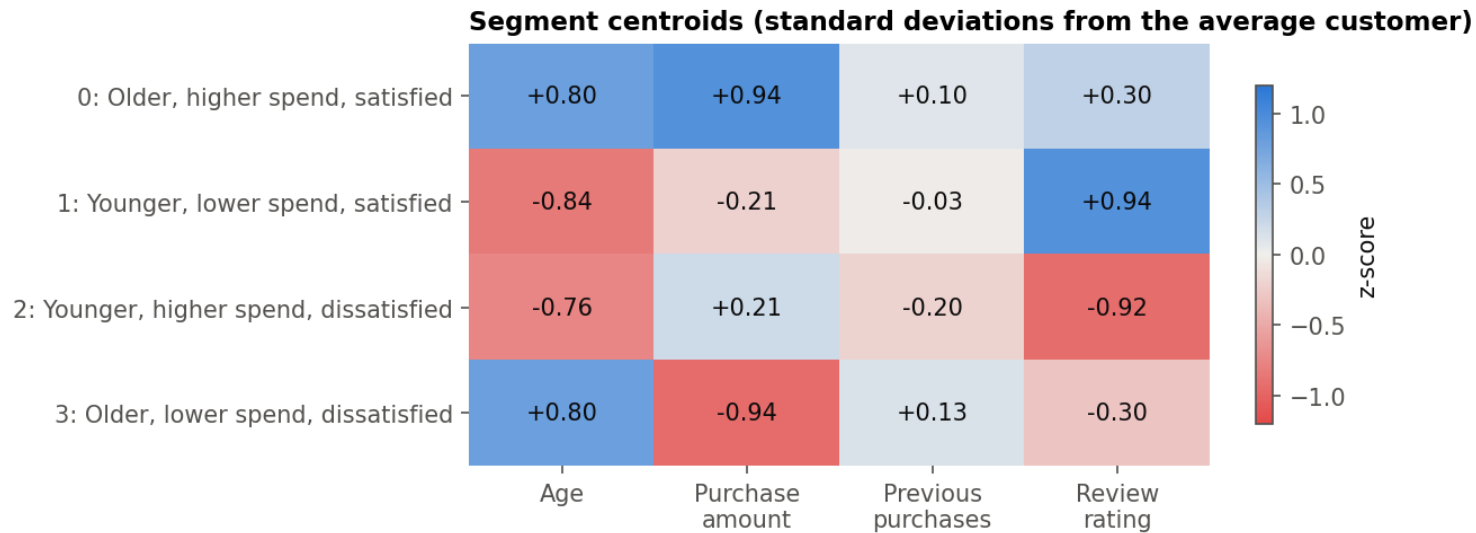
Stability across 10 seeds (ARI), and differences on variables not used for clustering

The silhouette score matches random noise at every k



No elbow, and no silhouette advantage over noise, so there is no natural cluster count. I chose **k = 4** because four segments are easy to act on.

Four segments: age x spend x satisfaction



0 Older, higher spend, satisfied

Age 56 | Spend \$82 | Rating 4.0 | n = 966

1 Younger, lower spend, satisfied

Age 31 | Spend \$55 | Rating 4.4 | n = 969

2 Younger, higher spend, dissatisfied

Age 33 | Spend \$65 | Rating 3.1 | n = 986

3 Older, lower spend, dissatisfied

Age 56 | Spend \$38 | Rating 3.5 | n = 979

Previous purchases hardly differs between segments. The split is driven by age, spend and rating.

What to do with each segment

0 Older, higher spend, satisfied

Protect and reward: VIP perks, early access, referral incentives

1 Younger, lower spend, satisfied

Grow spend: bundles, cross-sell, personalised upsell offers

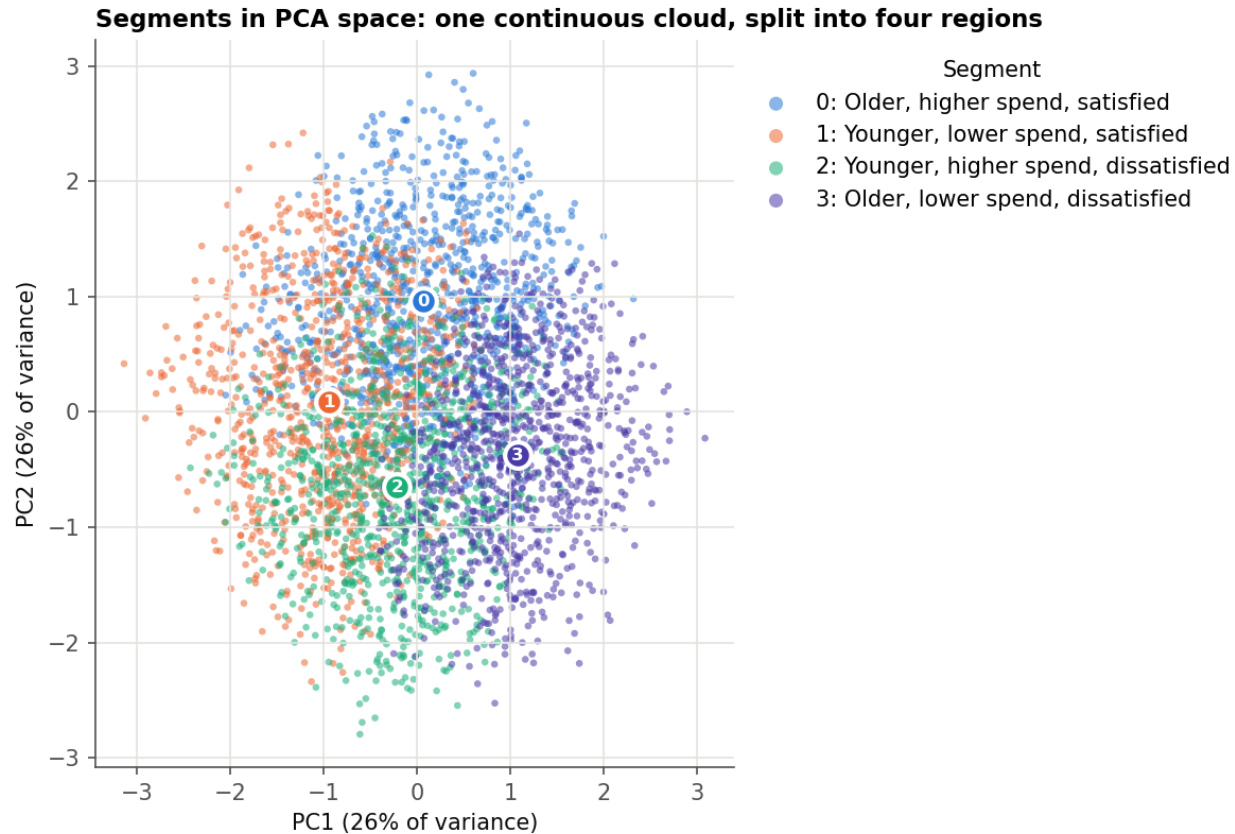
2 Younger, higher spend, dissatisfied

Most at-risk revenue: follow up on reviews, service recovery

3 Older, lower spend, dissatisfied

Re-engage cheaply: feedback surveys, targeted win-back offers

Are the segments real? Not really



One continuous cloud, divided into regions rather than separate groups.

- **Silhouette** = noise at every k
- **Unstable**: median ARI 0.47 across random seeds
- **No external difference**: subscription 26-28%, discounts 42-45%, same category mix

K-Means always returns k clusters, even when the data has none. The noise baseline catches that.

Limitations

- **The dataset is small.** 3,900 customers, each with a single transaction and no dates, so behaviour over time can't be studied. Cross-tabs get thin fast (about 39 purchases per item-season cell), which limits the power to detect small effects, and the findings may not generalise to a larger or real customer base.
- **The data is synthetic.** Its features are independent and evenly spread, so no clustering method could find natural groups in it. The pipeline is ready to run on real transaction data.
- **Categorical features are left out.** K-Means needs numeric inputs, so category, season and payment method aren't used. K-Prototypes or Gower distance with hierarchical clustering could include them.
- **Fairness.** Segments based on age should be reviewed before use, so that offers don't systematically exclude groups that are under-represented in the data.

Takeaways and next steps

- Use the four segments as a **simple targeting grid**, starting with segment 2 (younger, higher spend, dissatisfied), which has the most revenue at risk
- A plain rule on median age, spend and rating would work just as well on this data
- Don't target by gender or subscription alone, since neither predicts behaviour here
- A/B test segment offers before scaling them
- Next: real, larger, multi-period data, RFM features, and K-Prototypes for categorical columns